

AERIS Expert Review Packet

Anomaly

Source / metric	openaq / PM10 (pm10)
Observed / expected baseline	87 / 22.6522 ug/m3
Baseline method	mean of the preceding 7 days of this series (Z-score detector)
Time	2026-06-07 05:00 UTC
Location	29.7690, -95.3179
Severity	severe
Detectors	zscore, stl, isolation_forest

Stored evidence summary

The three tables in this section are look-up tables. You never have to read them through: they are here for the moment a claim quotes a number and you want to check it.

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
asos	Relative humidity (humidity)	percent	9 / 640	61.34 to 100; mean 85.4038	94.29 at 2026-06-07 04:55Z; 5 min before event
asos	Air temperature (temperature)	degC	9 / 640	23.8889 to 31.6667; mean 27.2583	27 at 2026-06-07 04:55Z; 5 min before event
asos	Wind direction (wind_direction)	degree	9 / 640	0 to 360; mean 139.922	160 at 2026-06-07 04:55Z; 5 min before event
asos	Wind speed (wind_speed)	m/s	9 / 640	0 to 10.8033; mean 3.6333	5.65888 at 2026-06-07 04:55Z; 5 min before event
noaa_gfs	500-hPa geopotential height (gh_500)	m	11 / 132	5853.96 to 5892.69; mean 5871.17	5868.98 at 2026-06-07 06:00Z; 1 h after event
noaa_gfs	Planetary boundary layer height (pbl_height)	m	11 / 132	44.2979 to 1425.04; mean 610.514	491.034 at 2026-06-07 06:00Z; 1 h after event
noaa_gfs	Precipitable water (precipitable_water)	mm	11 / 132	47.8843 to 59.8974; mean 55.3793	56.2208 at 2026-06-07 06:00Z; 1 h after event
noaa_gfs	Surface pressure (surface_pressure)	hPa	11 / 132	1001.91 to 1012.88; mean 1008.07	1008.52 at 2026-06-07 06:00Z; 1 h after event
noaa_gfs	850-hPa temperature (t_850)	degC	11 / 132	17.231 to 20.0187; mean 18.6295	18.9874 at 2026-06-07 06:00Z; 1 h after event
noaa_gfs	10-m eastward wind (u_10m)	m/s	11 / 132	-4.43892 to 1.30517; mean -1.2812	-0.910293 at 2026-06-07 06:00Z; 1 h after event
noaa_gfs	10-m northward wind (v_10m)	m/s	11 / 132	0.345771 to 6.69698; mean 3.0687	3.00933 at 2026-06-07 06:00Z; 1 h after event
openaq	Ozone (ozone)	ppm	18 / 1229	0 to 0.038; mean 0.017	0.01 at 2026-06-07 05:00Z; at event time

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
openaq	PM10 (pm10)	ug/m3	3 / 205	6 to 87; mean 32.3707	87 at 2026-06-07 05:00Z; at event time
openaq	PM2.5 (pm25)	ug/m3	88 / 14131	-4.9 to 177.77; mean 6.922	25.4 at 2026-06-07 05:00Z; at event time
openweather	Cloud cover (cloud_cover)	percent	5 / 360	76 to 100; mean 98.3333	81 at 2026-06-07 04:57Z; 2 min before event
openweather	Relative humidity (humidity)	percent	5 / 360	70 to 96; mean 86.6389	95 at 2026-06-07 04:57Z; 2 min before event
openweather	Precipitation rate (precipitation)	mm/h	5 / 360	0 to 8.65; mean 0.0914	0 at 2026-06-07 04:57Z; 2 min before event
openweather	Surface pressure (pressure)	hPa	5 / 360	1008 to 1015; mean 1010.84	1010 at 2026-06-07 04:57Z; 2 min before event
openweather	Air temperature (temperature)	degC	5 / 360	24.44 to 31.48; mean 27.4367	26.62 at 2026-06-07 04:57Z; 2 min before event
openweather	Wind direction (wind_direction)	degree	5 / 360	0 to 355; mean 151.578	166 at 2026-06-07 04:57Z; 2 min before event
openweather	Wind speed (wind_speed)	m/s	5 / 360	0.45 to 5.94; mean 2.2153	3.67 at 2026-06-07 04:57Z; 2 min before event
purpleair	PM2.5 (pm25)	ug/m3	66 / 4771	0 to 286.4; mean 6.3857	6 at 2026-06-07 05:00Z; at event time
sentinel5p	Aerosol index granules (s5p_aer_ai_granule_available)	count	9 / 9	1 to 1; mean 1	1 at 2026-06-06 20:14Z; 8 h 45 min before event
sentinel5p	CH4 granules available (s5p_ch4_granule_available)	count	9 / 9	1 to 1; mean 1	1 at 2026-06-06 20:14Z; 8 h 45 min before event
sentinel5p	CO column (s5p_co_column)	mol/m^2	2 / 2	0.0279382 to 0.0279482; mean 0.0279	0.0279482 at 2026-06-06 20:14Z; 8 h 45 min before event
sentinel5p	CO granules available (s5p_co_granule_available)	count	9 / 9	1 to 1; mean 1	1 at 2026-06-06 20:14Z; 8 h 45 min before event
sentinel5p	HCHO column (s5p_hcho_column)	mol/m^2	4 / 4	8.2537e-05 to 0.00012874; mean 0.0001	0.00012874 at 2026-06-06 20:14Z; 8 h 45 min before event
sentinel5p	HCHO granules available (s5p_hcho_granule_available)	count	9 / 9	1 to 1; mean 1	1 at 2026-06-06 20:14Z; 8 h 45 min before event
sentinel5p	NO2 column (s5p_no2_column)	mol/m^2	1 / 1	3.15424e-05 to 3.15424e-05; mean 0	3.15424e-05 at 2026-06-06 20:14Z; 8 h 45 min before event
sentinel5p	NO2 granules available (s5p_no2_granule_available)	count	7 / 7	1 to 1; mean 1	1 at 2026-06-06 20:14Z; 8 h 45 min before event
sentinel5p	O3 granules available (s5p_o3_granule_available)	count	9 / 9	1 to 1; mean 1	1 at 2026-06-06 20:14Z; 8 h 45 min before event
sentinel5p	SO2 column (s5p_so2_column)	mol/m^2	4 / 4	-9.7923e-05 to 1.0299e-05; mean -0	-9.7923e-05 at 2026-06-06 20:14Z; 8 h 45 min before event
sentinel5p	SO2 granules available (s5p_so2_granule_available)	count	9 / 9	1 to 1; mean 1	1 at 2026-06-06 20:14Z; 8 h 45 min before event
tceq	Carbon monoxide (co)	ppm	3 / 215	0.1 to 0.8; mean 0.2233	0.1 at 2026-06-07 05:00Z; at event time
tceq	Nitrogen dioxide (no2)	ppb	13 / 908	-2.7 to 28.3; mean 3.5207	3.5 at 2026-06-07 05:00Z; at event time

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
tceq	Sulfur dioxide (so2)	ppb	3 / 212	-0.4 to 0.7; mean -0.1415	-0.2 at 2026-06-07 05:00Z; at event time

Six-hour averages

Each metric averaged in 6-hour blocks across the stored 72 hours, in UTC. These averages, and the per-site ones below, are what the models were shown.

Source	Metric	Values
asos	Relative humidity (humidity)	06-05 12Z 85.28, 18Z 81.6, 06-06 00Z 86.41, 06Z 93.76, 12Z 83.45, 18Z 82.93, 06-07 00Z 90.42, 06Z 92.7, 12Z 78.11, 18Z 73.18, 06-08 00Z 87.43, 06Z 91.28, 12Z 83.4
asos	Air temperature (temperature)	06-05 12Z 27.17, 18Z 27.9, 06-06 00Z 26.2, 06Z 24.98, 12Z 27.9, 18Z 27.62, 06-07 00Z 27.02, 06Z 26.41, 12Z 28.25, 18Z 28.63, 06-08 00Z 27.13, 06Z 26.81, 12Z 28.36
asos	Wind direction (wind_direction)	06-05 12Z 120, 18Z 114.4, 06-06 00Z 109.4, 06Z 83.14, 12Z 131.5, 18Z 154.2, 06-07 00Z 150.4, 06Z 146.7, 12Z 162.5, 18Z 164.8, 06-08 00Z 145.4, 06Z 158.3, 12Z 163.1
asos	Wind speed (wind_speed)	06-05 12Z 3.773, 18Z 4.64, 06-06 00Z 3.164, 06Z 1.574, 12Z 2.267, 18Z 3.294, 06-07 00Z 3.468, 06Z 3.02, 12Z 5.026, 18Z 5.011, 06-08 00Z 3.639, 06Z 3.915, 12Z 4.641
noaa_gfs	500-hPa geopotential height (gh_500)	06-05 18Z 5876, 06-06 00Z 5865, 06Z 5865, 12Z 5857, 18Z 5870, 06-07 00Z 5864, 06Z 5869, 12Z 5862, 18Z 5873, 06-08 00Z 5874, 06Z 5889, 12Z 5891
noaa_gfs	Planetary boundary layer height (pbl_height)	06-05 18Z 960.8, 06-06 00Z 718.5, 06Z 261.9, 12Z 133.3, 18Z 1103, 06-07 00Z 436.8, 06Z 447.2, 12Z 490.8, 18Z 1191, 06-08 00Z 619, 06Z 445.3, 12Z 518.4
noaa_gfs	Precipitable water (precipitable_water)	06-05 18Z 55.15, 06-06 00Z 52.45, 06Z 54.45, 12Z 55.66, 18Z 57.13, 06-07 00Z 57.97, 06Z 54.76, 12Z 56.04, 18Z 53.23, 06-08 00Z 54.27, 06Z 57.38, 12Z 56.08
noaa_gfs	Surface pressure (surface_pressure)	06-05 18Z 1010, 06-06 00Z 1008, 06Z 1008, 12Z 1008, 18Z 1008, 06-07 00Z 1006, 06Z 1007, 12Z 1007, 18Z 1008, 06-08 00Z 1007, 06Z 1009, 12Z 1010
noaa_gfs	850-hPa temperature (t_850)	06-05 18Z 17.39, 06-06 00Z 18.15, 06Z 17.94, 12Z 17.76, 18Z 18.3, 06-07 00Z 19.07, 06Z 19.17, 12Z 18.49, 18Z 19.11, 06-08 00Z 19.4, 06Z 19.54, 12Z 19.23
noaa_gfs	10-m eastward wind (u_10m)	06-05 18Z -2.739, 06-06 00Z -3.568, 06Z -2.025, 12Z -0.6111, 18Z 0.6088, 06-07 00Z -1.048, 06Z -0.9667, 12Z -0.7284, 18Z -1.019, 06-08 00Z -0.9179, 06Z -1.23, 12Z -1.131
noaa_gfs	10-m northward wind (v_10m)	06-05 18Z 2.402, 06-06 00Z 2.35, 06Z 1.535, 12Z 1.625, 18Z 2.687, 06-07 00Z 3.449, 06Z 3.354, 12Z 3.478, 18Z 5.579, 06-08 00Z 3.696, 06Z 2.952, 12Z 3.718
openaq	Ozone (ozone)	06-05 12Z 0.02564, 18Z 0.02563, 06-06 00Z 0.01893, 06Z 0.01189, 12Z 0.01376, 18Z 0.02306, 06-07 00Z 0.01591, 06Z 0.01245, 12Z 0.01811, 18Z 0.02124, 06-08 00Z 0.01469, 06Z 0.01534, 12Z 0.01342
openaq	PM10 (pm10)	06-05 12Z 12, 18Z 11.91, 06-06 00Z 20.08, 06Z 13.44, 12Z 19.39, 18Z 13.17, 06-07 00Z 40.83, 06Z 51.06, 12Z 43.61, 18Z 41.22, 06-08 00Z 40.94, 06Z 39.5, 12Z 43.5
openaq	PM2.5 (pm25)	06-05 12Z 4.161, 18Z 3.807, 06-06 00Z 2.954, 06Z 3.064, 12Z 4.436, 18Z 4.081, 06-07 00Z 7.865, 06Z 9.41, 12Z 7.992, 18Z 6.984, 06-08 00Z 8.275, 06Z 8.576, 12Z 8.048
openweather	Cloud cover (cloud_cover)	06-05 12Z 87.6, 18Z 99.37, 06-06 00Z 98.83, 06Z 100, 12Z 99.9, 18Z 94.5, 06-07 00Z 93.43, 06Z 98.77, 12Z 97.8, 18Z 100, 06-08 00Z 99.53, 06Z 99.93, 12Z 100
openweather	Relative humidity (humidity)	06-05 12Z 85.4, 18Z 84.93, 06-06 00Z 87.57, 06Z 92.7, 12Z 85.87, 18Z 82.93, 06-07 00Z 90.23, 06Z 92.57, 12Z 82.47, 18Z 77, 06-08 00Z 87.13, 06Z 91.3, 12Z 84.88
openweather	Precipitation rate (precipitation)	06-05 12Z 0.364, 18Z 0.4553, 06-06 00Z 0, 06Z 0, 12Z 0.1037, 18Z 0.3413, 06-07 00Z 0, 06Z 0, 12Z 0, 18Z 0.003667, 06-08 00Z 0, 06Z 0.08733, 12Z 0.054
openweather	Surface pressure (pressure)	06-05 12Z 1013, 18Z 1011, 06-06 00Z 1011, 06Z 1010, 12Z 1011, 18Z 1009, 06-07 00Z 1010, 06Z 1009, 12Z 1011, 18Z 1010, 06-08 00Z 1011, 06Z 1012, 12Z 1014
openweather	Air temperature (temperature)	06-05 12Z 27.7, 18Z 28.08, 06-06 00Z 26.3, 06Z 24.96, 12Z 27.96, 18Z 28.49, 06-07 00Z 27.14, 06Z 26.34, 12Z 28.42, 18Z 28.97, 06-08 00Z 27.36, 06Z 26.87, 12Z 28.48
openweather	Wind direction (wind_direction)	06-05 12Z 155.4, 18Z 116.5, 06-06 00Z 116.8, 06Z 136.8, 12Z 174.1, 18Z 145.4, 06-07 00Z 153.1, 06Z 159.3, 12Z 163.5, 18Z 176.2, 06-08 00Z 154, 06Z 163.4, 12Z 160.7
openweather	Wind speed (wind_speed)	06-05 12Z 2.782, 18Z 2.533, 06-06 00Z 2.195, 06Z 1.636, 12Z 1.706, 18Z 2.441, 06-07 00Z 1.969, 06Z 1.796, 12Z 2.877, 18Z 2.716, 06-08 00Z 1.899, 06Z 2.159, 12Z 2.631

Source	Metric	Values
purpleair	PM2.5 (pm25)	06-05 12Z 2.203, 18Z 2.405, 06-06 00Z 2.021, 06Z 1.871, 12Z 3.853, 18Z 3.907, 06-07 00Z 9.549, 06Z 10.39, 12Z 7.687, 18Z 6.841, 06-08 00Z 9.869, 06Z 10.17, 12Z 8.726
sentinel5p	Aerosol index granules (s5p_aer_ai_granule_available)	06-05 18Z 1, 06-06 18Z 1, 06-07 18Z 1
sentinel5p	CH4 granules available (s5p_ch4_granule_available)	06-05 18Z 1, 06-06 18Z 1, 06-07 18Z 1
sentinel5p	CO granules available (s5p_co_granule_available)	06-05 18Z 1, 06-06 18Z 1, 06-07 18Z 1
sentinel5p	HCHO column (s5p_hcho_column)	06-05 18Z 8.291e-05, 06-06 18Z 0.0001235
sentinel5p	HCHO granules available (s5p_hcho_granule_available)	06-05 18Z 1, 06-06 18Z 1, 06-07 18Z 1
sentinel5p	NO2 granules available (s5p_no2_granule_available)	06-05 18Z 1, 06-06 18Z 1, 06-07 18Z 1
sentinel5p	O3 granules available (s5p_o3_granule_available)	06-05 18Z 1, 06-06 18Z 1, 06-07 18Z 1
sentinel5p	SO2 column (s5p_so2_column)	06-05 18Z 5.139e-06, 06-06 18Z -9.483e-05
sentinel5p	SO2 granules available (s5p_so2_granule_available)	06-05 18Z 1, 06-06 18Z 1, 06-07 18Z 1
tceq	Carbon monoxide (co)	06-05 12Z 0.1667, 18Z 0.2, 06-06 00Z 0.2889, 06Z 0.2, 12Z 0.2722, 18Z 0.2778, 06-07 00Z 0.2778, 06Z 0.1333, 12Z 0.1722, 18Z 0.1933, 06-08 00Z 0.2389, 06Z 0.1667, 12Z 0.2611
tceq	Nitrogen dioxide (no2)	06-05 12Z 5.462, 18Z 5.435, 06-06 00Z 5.392, 06Z 5.67, 12Z 3.539, 18Z 5.16, 06-07 00Z 3.372, 06Z 1.548, 12Z 1.372, 18Z 1.385, 06-08 00Z 2.668, 06Z 2.06, 12Z 3.814
tceq	Sulfur dioxide (so2)	06-05 12Z -0.1333, 18Z -0.07222, 06-06 00Z -0.05, 06Z -0.1333, 12Z -0.1056, 18Z 0.04444, 06-07 00Z -0.1714, 06Z -0.2375, 12Z -0.2278, 18Z -0.1833, 06-08 00Z -0.1722, 06Z -0.25, 12Z -0.1588

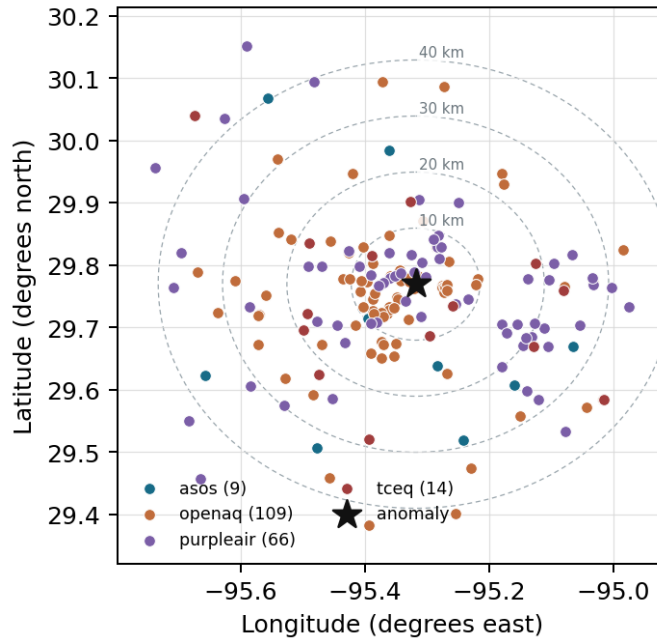
Per-site averages

Each metric averaged at each site, with that site's distance from the event in brackets.

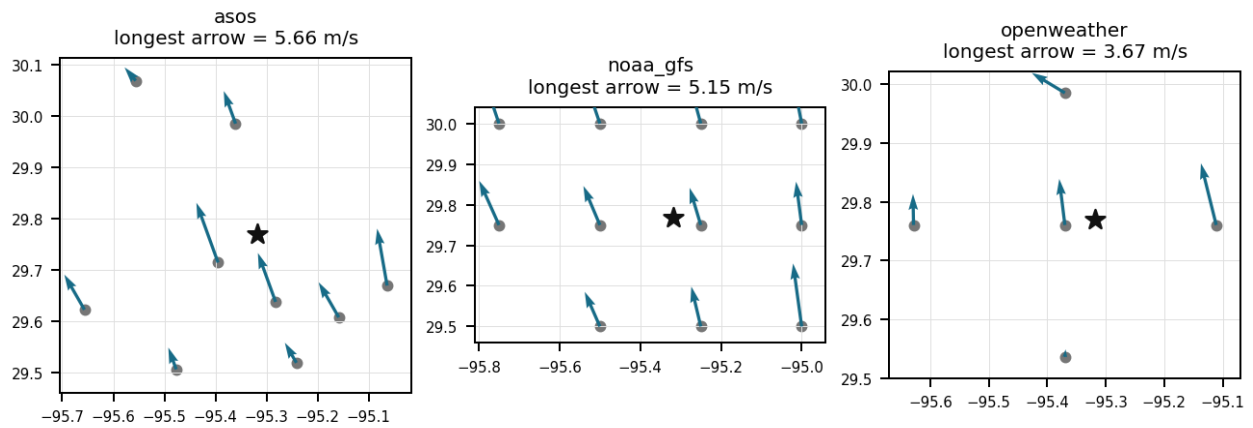
Source	Metric	Values
asos	Relative humidity (humidity)	MCJ (9.638 km) 89.01, HOU (15.02 km) 84.92, EFD (23.659 km) 86.63, IAH (24.303 km) 86.13, T41 (26.902 km) 84.29, LVJ (28.769 km) 83.85, AXH (33.03 km) 81.44, SGR (36.553 km) 87.68, +1 more
asos	Air temperature (temperature)	MCJ (9.638 km) 26.87, HOU (15.02 km) 27.36, EFD (23.659 km) 27.52, IAH (24.303 km) 27.06, T41 (26.902 km) 27.67, LVJ (28.769 km) 27.47, AXH (33.03 km) 27.34, SGR (36.553 km) 27.15, +1 more
asos	Wind direction (wind_direction)	MCJ (9.638 km) 148.7, HOU (15.02 km) 156.4, EFD (23.659 km) 141.1, IAH (24.303 km) 141.1, T41 (26.902 km) 161.2, LVJ (28.769 km) 131.5, AXH (33.03 km) 115, SGR (36.553 km) 146, +1 more
asos	Wind speed (wind_speed)	MCJ (9.638 km) 5.391, HOU (15.02 km) 3.987, EFD (23.659 km) 4.503, IAH (24.303 km) 3.58, T41 (26.902 km) 4.616, LVJ (28.769 km) 2.979, AXH (33.03 km) 1.815, SGR (36.553 km) 3.751, +1 more
noaa_gfs	500-hPa geopotential height (gh_500)	gfs:29.75,-95.25 (6.883 km) 5872, gfs:29.75,-95.50 (17.709 km) 5870, gfs:30.00,-95.25 (26.504 km) 5871, gfs:29.50,-95.25 (30.624 km) 5873, gfs:29.75,-95.00 (30.756 km) 5873, gfs:30.00,-95.50 (31.113 km) 5869, gfs:29.50,-95.50 (34.709 km) 5872, gfs:30.00,-95.00 (39.984 km) 5872, +3 more
noaa_gfs	Planetary boundary layer height (pbl_height)	gfs:29.75,-95.25 (6.883 km) 671.7, gfs:29.75,-95.50 (17.709 km) 658.9, gfs:30.00,-95.25 (26.504 km) 675.3, gfs:29.50,-95.25 (30.624 km) 579.5, gfs:29.75,-95.00 (30.756 km) 611, gfs:30.00,-95.50 (31.113 km) 694, gfs:29.50,-95.50 (34.709 km) 518.3, gfs:30.00,-95.00 (39.984 km) 521.5, +3 more

Source	Metric	Values
noaa_gfs	Precipitable water (precipitable_water)	gfs:29.75,-95.25 (6.883 km) 55.66, gfs:29.75,-95.50 (17.709 km) 55.64, gfs:30.00,-95.25 (26.504 km) 55.39, gfs:29.50,-95.25 (30.624 km) 55.5, gfs:29.75,-95.00 (30.756 km) 55.3, gfs:30.00,-95.50 (31.113 km) 55.25, gfs:29.50,-95.50 (34.709 km) 55.75, gfs:30.00,-95.00 (39.984 km) 55.43, +3 more
noaa_gfs	Surface pressure (surface_pressure)	gfs:29.75,-95.25 (6.883 km) 1009, gfs:29.75,-95.50 (17.709 km) 1008, gfs:30.00,-95.25 (26.504 km) 1008, gfs:29.50,-95.25 (30.624 km) 1009, gfs:29.75,-95.00 (30.756 km) 1010, gfs:30.00,-95.50 (31.113 km) 1006, gfs:29.50,-95.50 (34.709 km) 1008, gfs:30.00,-95.00 (39.984 km) 1009, +3 more
noaa_gfs	850-hPa temperature (t_850)	gfs:29.75,-95.25 (6.883 km) 18.6, gfs:29.75,-95.50 (17.709 km) 18.67, gfs:30.00,-95.25 (26.504 km) 18.57, gfs:29.50,-95.25 (30.624 km) 18.68, gfs:29.75,-95.00 (30.756 km) 18.61, gfs:30.00,-95.50 (31.113 km) 18.65, gfs:29.50,-95.50 (34.709 km) 18.67, gfs:30.00,-95.00 (39.984 km) 18.58, +3 more
noaa_gfs	10-m eastward wind (u_10m)	gfs:29.75,-95.25 (6.883 km) -1.219, gfs:29.75,-95.50 (17.709 km) -1.163, gfs:30.00,-95.25 (26.504 km) -1.137, gfs:29.50,-95.25 (30.624 km) -1.478, gfs:29.75,-95.00 (30.756 km) -1.306, gfs:30.00,-95.50 (31.113 km) -1.14, gfs:29.50,-95.50 (34.709 km) -1.419, gfs:30.00,-95.00 (39.984 km) -1.233, +3 more
noaa_gfs	10-m northward wind (v_10m)	gfs:29.75,-95.25 (6.883 km) 2.876, gfs:29.75,-95.50 (17.709 km) 2.818, gfs:30.00,-95.25 (26.504 km) 2.679, gfs:29.50,-95.25 (30.624 km) 3.175, gfs:29.75,-95.00 (30.756 km) 3.255, gfs:30.00,-95.50 (31.113 km) 2.705, gfs:29.50,-95.50 (34.709 km) 2.86, gfs:30.00,-95.00 (39.984 km) 2.946, +3 more
openaq	Ozone (ozone)	3378 (7.015 km) 0.01614, 2046 (7.328 km) 0.01318, 1319333 (7.534 km) 0.01883, 2039 (9.388 km) 0.01511, 325 (9.456 km) 0.01621, 310 (14.708 km) 0.0175, 320 (16.681 km) 0.01374, 319 (18.051 km) 0.01547, +10 more
openaq	PM10 (pm10)	15852419 (0.0 km) 41.17, 13380082 (7.015 km) 29.34, 271 (21.343 km) 28.11
openaq	PM2.5 (pm25)	15852418 (0.0 km) 15.99, 15588191 (0.889 km) 7.219, 8967237 (1.063 km) 3.676, 12256170 (1.277 km) 7.514, 8967273 (1.528 km) 5.663, 8967217 (1.567 km) 5.878, 8967005 (1.748 km) 8.225, 15596557 (2.479 km) 6.907, +80 more
openweather	Cloud cover (cloud_cover)	grid:center (5.104 km) 99.29, grid:east (19.981 km) 97.18, grid:north (24.532 km) 99.26, grid:south (26.414 km) 97.03, grid:west (30.0 km) 98.9
openweather	Relative humidity (humidity)	grid:center (5.104 km) 86.36, grid:east (19.981 km) 89.18, grid:north (24.532 km) 86.64, grid:south (26.414 km) 84.81, grid:west (30.0 km) 86.21
openweather	Precipitation rate (precipitation)	grid:center (5.104 km) 0.1031, grid:east (19.981 km) 0.02944, grid:north (24.532 km) 0.1486, grid:south (26.414 km) 0.1351, grid:west (30.0 km) 0.04083
openweather	Surface pressure (pressure)	grid:center (5.104 km) 1011, grid:east (19.981 km) 1011, grid:north (24.532 km) 1011, grid:south (26.414 km) 1011, grid:west (30.0 km) 1011
openweather	Air temperature (temperature)	grid:center (5.104 km) 27.54, grid:east (19.981 km) 27.34, grid:north (24.532 km) 27.27, grid:south (26.414 km) 27.62, grid:west (30.0 km) 27.41
openweather	Wind direction (wind_direction)	grid:center (5.104 km) 148.6, grid:east (19.981 km) 159.8, grid:north (24.532 km) 132.1, grid:south (26.414 km) 164.2, grid:west (30.0 km) 153.3
openweather	Wind speed (wind_speed)	grid:center (5.104 km) 2.157, grid:east (19.981 km) 3.564, grid:north (24.532 km) 1.939, grid:south (26.414 km) 1.968, grid:west (30.0 km) 1.449
purpleair	PM2.5 (pm25)	222601 (2.006 km) 3.648, 229853 (2.189 km) 3.397, 308976 (2.99 km) 6.459, 133994 (3.421 km) 0.3959, 186303 (3.504 km) 7.267, 235427 (3.982 km) 6.292, 127451 (4.154 km) 2.114, 27821 (5.042 km) 0.2932, +58 more
tceq	Carbon monoxide (co)	482011035 (7.017 km) 0.1451, 482011052 (8.428 km) 0.3889, 482011039 (21.358 km) 0.1347
tceq	Nitrogen dioxide (no2)	482011035 (7.017 km) 5.673, 482011052 (8.428 km) 7.241, 482010416 (9.465 km) 4.015, 482010024 (14.701 km) 0.58, 482011066 (17.679 km) 4.454, 482010047 (18.045 km) 4.886, 482010026 (18.937 km) 6.114, 482010055 (19.314 km) 2.459, +5 more
tceq	Sulfur dioxide (so2)	482011035 (7.017 km) -0.2319, 482011039 (21.358 km) 0.07391, 482010051 (22.104 km) -0.2592

Ground observation locations in the stored 72-hour context
dashed rings are great-circle distance from the anomaly



Event-nearest wind vectors from stored values (arrow points downwind)



Claims

Mark exactly one box per claim: V (Valid), I (Invalid), or U (Unsure). To skip a claim, leave all three boxes blank; a blank is recorded as missing and never turned into Unsure. Each claim appears exactly once, and the number printed next to it is how I'll refer to that claim if I need to ask you about it.

Label definitions, from the labeling guide

Valid (V): the claim holds up at the precision it states, and the evidence in the packet supports it.

Invalid (I): the evidence in the packet contradicts the claim, the claim misstates a measurement, or the physical reasoning does not hold.

Unsure (U): you cannot tell from what is shown. This covers thin evidence, measurements that cannot resolve the claim, ambiguous wording, and anything outside what you are comfortable judging.

Missing or thin evidence means Unsure, not Invalid.

A cause that is plausible but unproven stays Unsure unless the evidence actually contradicts it.

Some claims bundle several statements into one sentence. Judge those by their parts. If every part earns the same verdict, use that verdict.

Claim 1 of 36

Marine aerosol or salt-rich coarse particle transport from the Gulf is a plausible contributor to the Houston PM10 anomaly because surface winds were consistently from the southeast to south-southeast near 150 to 166 degrees during a very humid period, indicating sustained onshore flow into the urban area.

Mark exactly one:

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Note (optional):

Claim 2 of 36

High ambient relative humidity exceeding 92 percent near 05:00 UTC on June 7, 2026, promoted hygroscopic growth of atmospheric aerosols, increasing measured PM10 mass.

Mark exactly one:

☐ V

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Note (optional):

Claim 3 of 36

The air quality anomaly in Houston, Texas is caused by high levels of CO.

Mark exactly one:

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Note (optional):

Claim 4 of 36

High relative humidity recorded by asos and openweather near 94 percent facilitated hygroscopic particle growth, increasing aerosol mass concentrations measured by openaq sensors.

Mark exactly one:

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Note (optional):

Claim 5 of 36

At 2026-06-07T05:00:00+00:00 near the anomaly location, openaq PM2.5 was 25.4 ug/m3, openaq ozone was 0.01 ppm, tceq CO was 0.1 ppm, tceq NO2 was 3.5 ppb, and tceq SO2 was -0.2 ppb.

Mark exactly one:

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Note (optional):

Claim 6 of 36

The air quality anomaly in Houston, Texas is characterized by elevated levels of PM2.5.

Mark exactly one:

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Note (optional):

Claim 7 of 36

A local coarse-particle source such as road dust, construction material, industrial bulk handling, or other mechanically generated dust is supported because openaq PM10 at the anomaly site rose to 87 ug/m3 while PM2.5 at the same site was only 25.4 ug/m3 and regional PM2.5 was much lower near 6 ug/m3, which indicates a particle-size distribution weighted toward coarse material rather than a broadly mixed fine-particle episode.

Mark exactly one:☐ V☐ I☐ U**Note (optional):****Claim 8 of 36**

ASOS surface observations recorded high relative humidity above 90 percent and persistent southerly winds during the anomaly, promoting hygroscopic growth of suspended particulates.

Mark exactly one:☐ V☐ I☐ U**Note (optional):****Claim 9 of 36**

The wind direction at the time of the anomaly was around 166.0 degrees.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 10 of 36

Regional 6-hour average PM10 concentrations measured by openaq sensors increased from 13.17 ug/m3 on June 6 to a peak of 51.06 ug/m3 during the 06-07 06Z window.

Mark exactly one:☐ V☐ I☐ U**Note (optional):****Claim 11 of 36**

The air quality anomaly in Houston, Texas is caused by high levels of PM2.5.

Mark exactly one:☐ V☐ I☐ U**Note (optional):****Claim 12 of 36**

The temperature inversion caused by the high-pressure system over the Gulf of Mexico led to a stagnation of air, resulting in poor air quality.

Mark exactly one:☐ V☐ I☐ U**Note (optional):****Claim 13 of 36**

The air quality anomaly in Houston, Texas is likely caused by a combination of factors including industrial activities and vehicle emissions.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 14 of 36

A reduced nocturnal planetary boundary layer height of approximately 447 to 491 meters restricted vertical atmospheric dispersion, trapping particulate emissions near the ground.

Mark exactly one:☐ V☐ I☐ U**Note (optional):****Claim 15 of 36**

The presence of pollutants from industrial activities and vehicle emissions in the Houston area contributed to the poor air quality.

Mark exactly one:☐ V☐ I☐ U**Note (optional):****Claim 16 of 36**

The wind direction and speed, which were not favorable for dispersing pollutants, also played a role in the poor air quality.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 17 of 36

Resuspended road dust or construction-related dust is a leading hypothesis for the Houston PM10 anomaly because PM10 rose sharply to 87 $\mu\text{g}/\text{m}^3$ while nearby PM2.5 increased much less strongly and gaseous combustion tracers remained low, which matches a coarse-particle-dominant source signature.

Mark exactly one:

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Note (optional):

Claim 18 of 36

The temperature at the time of the anomaly was around 26.62°C.

Mark exactly one:

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Note (optional):

Claim 19 of 36

The Houston PM10 spike was likely inflated but not fully explained by humidity because local PM2.5 also increased while nearby PurpleAir PM2.5 showed a regional overnight rise.

Mark exactly one:

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Note (optional):

Claim 20 of 36

The air quality anomaly in Houston, Texas is caused by low wind speeds.

Mark exactly one:

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Note (optional):

Claim 21 of 36

The openaq PM10 6-hour mean increased from 13.17 ug/m3 during 2026-06-06 18Z to 40.83 ug/m3 during 2026-06-07 00Z and 51.06 ug/m3 during 2026-06-07 06Z, indicating that elevated PM10 persisted across multiple hours around the anomaly time.

Mark exactly one:

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Note (optional):

Claim 22 of 36

An OpenAQ ground sensor detected a severe PM10 concentration spike of 87.0 ug/m3 in Houston at 05:00 UTC on June 7, 2026, concurrent with increased fine particulate levels across the PurpleAir sensor network.

Mark exactly one:

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Note (optional):

Claim 23 of 36

Sustained southerly to southeasterly surface winds observed by asos and openweather transported regional coarse particulate matter into the Houston metropolitan area, elevating PM10 levels across openaq monitoring networks.

Mark exactly one:

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Note (optional):

Claim 24 of 36

A PM10 concentration anomaly of 87.0 ug/m³, against an expected value of approximately 22.65 ug/m³ (z-score 5.56), was detected at coordinates (29.769, -95.318) on June 7, 2026 at 05:00 UTC.

Mark exactly one:☐ V☐ I☐ U**Note (optional):****Claim 25 of 36**

Nighttime boundary-layer confinement likely amplified near-surface PM10 concentrations in Houston because the anomaly occurred under high humidity and a relatively shallow planetary boundary layer of about 491 m, which would favor accumulation of locally emitted or transported particles before daytime mixing increased.

Mark exactly one:☐ V☐ I☐ U**Note (optional):****Claim 26 of 36**

The air quality anomaly in Houston, Texas is also characterized by elevated levels of NO₂.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 27 of 36

Humidity-related measurement enhancement remains plausible but not sufficient as a sole explanation because the event occurred with very high relative humidity near 94 to 95 percent, cloud cover was extensive, and nighttime boundary-layer height was only about 491 m, yet the PM10 increase was much larger than the concurrent PM2.5 increase and nearby PurpleAir PM2.5 did not show a comparably extreme spike.

Mark exactly one:☐ V☐ I☐ U**Note (optional):****Claim 28 of 36**

At the exact time of the PM10 anomaly, the local openaq PM2.5 reading reached 25.4 ug/m3 at the same location, coinciding with elevated regional 6-hour mean PM2.5 levels of up to 10.39 ug/m3 reported by purpleair.

Mark exactly one:☐ V☐ I☐ U**Note (optional):****Claim 29 of 36**

The Houston meteorological setting near 2026-06-07T05:00:00+00:00 favored nighttime accumulation and transport of coarse particles because humidity was about 94 to 95 percent, winds were from the southeast at about 3.7 to 5.7 m/s, and modeled planetary boundary layer height was about 491 m.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 30 of 36

NOAA GFS reanalysis data showed a shallow nocturnal planetary boundary layer height dropping below 450 meters during the early morning of June 7, 2026, trapping particulates near the surface.

Mark exactly one:

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Note (optional):

Claim 31 of 36

Sustained southerly to southeasterly surface winds transported regional airborne particulate matter into Houston, elevating regional PM10 levels across multiple monitoring sensors.

Mark exactly one:

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Note (optional):

Claim 32 of 36

The openaq PM10 measurement at 29.769, -95.318 in Houston was 87.0 ug/m3 at 2026-06-07T05:00:00+00:00, compared with an expected value of 22.652173913043477 ug/m3, yielding a z-score of 5.564932313403645 and a severe anomaly classification.

Mark exactly one:

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Note (optional):

Claim 33 of 36

The Houston anomaly is most consistent with a local coarse-particle non-combustion event because PM10 reached 87 ug/m3 while ozone, NO2, CO, and SO2 stayed low near the anomaly time.

Mark exactly one:☐ V☐ I☐ U**Note (optional):****Claim 34 of 36**

Nocturnal boundary layer compression recorded by gfs and light surface winds observed by asos constrained atmospheric mixing, trapping ground-level coarse particulates and causing the openaq PM10 concentration spike.

Mark exactly one:☐ V☐ I☐ U**Note (optional):****Claim 35 of 36**

A combustion-dominated plume is contradicted because co-located or nearby combustion tracers remained low during the PM10 spike, with ozone near 0.01 ppm, TCEQ CO near 0.1 ppm, TCEQ NO2 near 3.5 ppb, and TCEQ SO2 near background or negative values, which is not the pattern expected for a strong fresh combustion event.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 36 of 36

The air quality anomaly is related to particulate matter (PM2.5) with a value of 6.0 ug/m3.

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Note (optional):

Optional: most likely cause

Your best guess at the true cause of this event, in your own words. "Insufficient evidence to determine a single cause" is a perfectly fine answer.